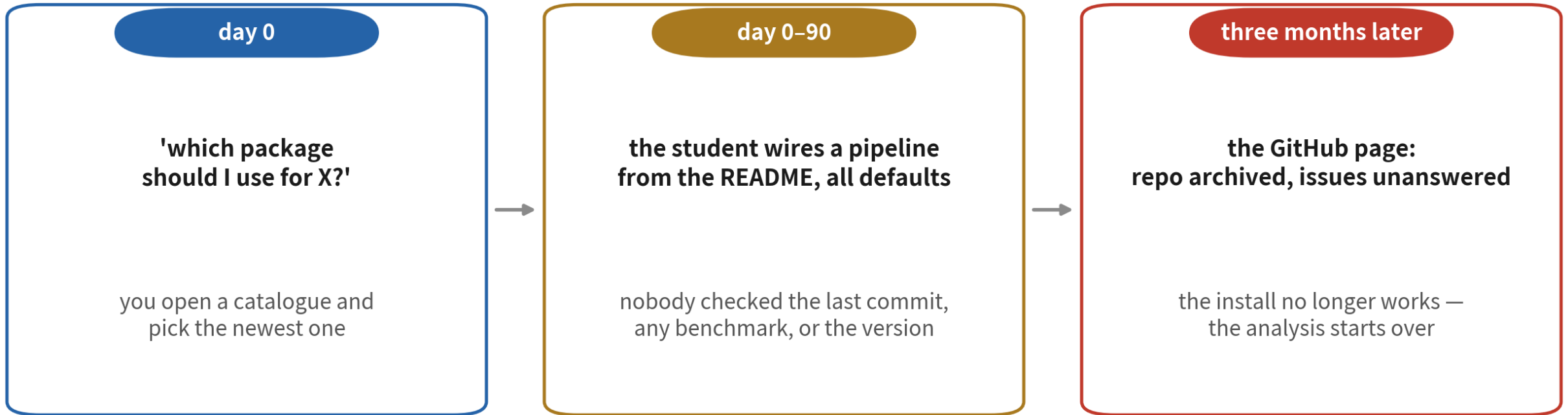


The tool landscape: choosing from eighteen hundred tools

Catalogues, benchmarks, official tutorials, version records — the four steps of tool choice, plus the R and Python ecosystem maps.

About 35 min | Prerequisite: B18 | Series B Supplement, Part 3

Three months later, the package was dead



the package is not the villain — the selection method is: newest in the catalogue, nothing checked in between

schematic

The package is not the villain — the selection method is: newest in the catalogue, nothing checked in between

WHERE THIS EPISODE STARTS

**You are not short of tools.
You are short of a method for choosing them.**

Single-cell tools number past eighteen hundred. A dying tool sends no notification — the method is your insurance.

What you will learn

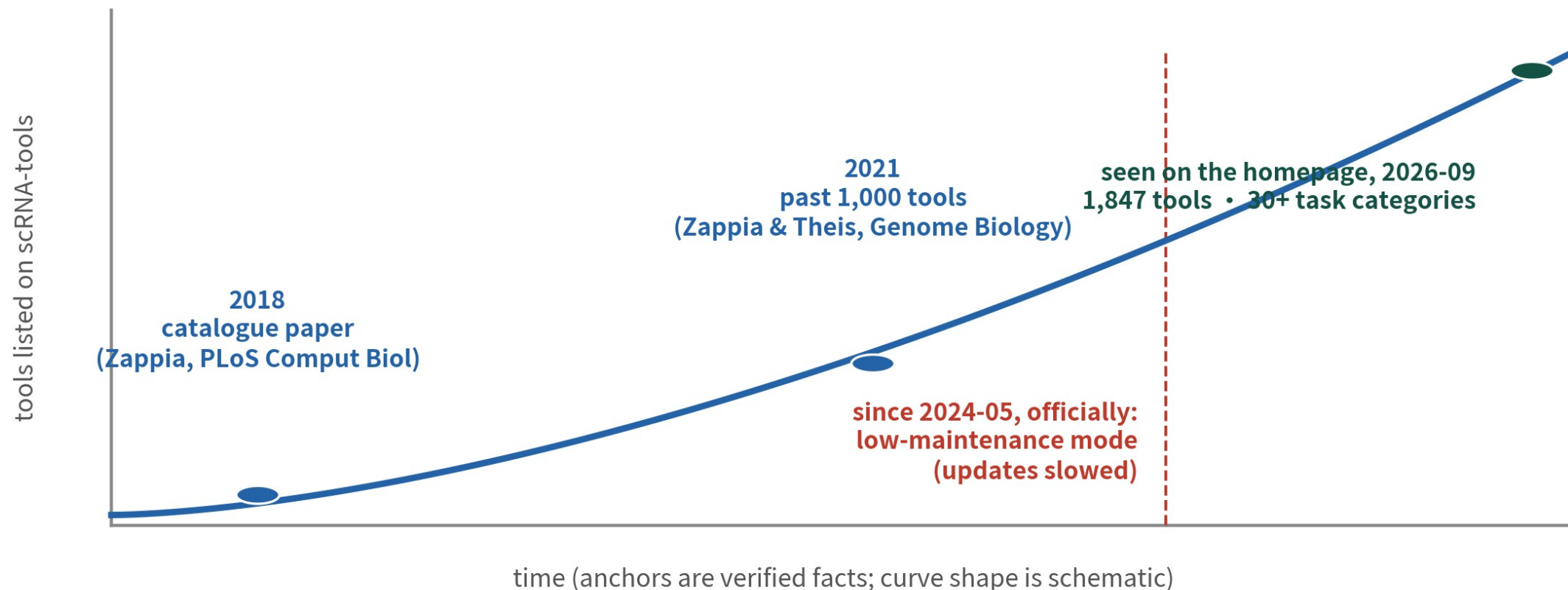
- 1 Choose a tool for any task in four steps: candidates from catalogues, evidence from benchmarks, confirm on the official tutorial, record the version
- 2 Read the task-by-task map of the R and Python ecosystems, and hand data across them with B18's checks
- 3 Pick the right learning resources for yourself and your students — which book, which course, and which dead ones to stop recommending

01

The tool explosion

How we got to eighteen hundred tools — and what a catalogue listing actually means

How we got to eighteen hundred tools



cite 1,847 with a date — the catalogue itself has slowed; for the newest tools, use awesome-single-cell

schematic

The anchors are verified facts; the curve's shape is schematic — cite 1,847 with a date

scRNA-tools today, in three numbers

All seen on the official pages, 2026-09

1,847

tools listed (seen on the homepage)

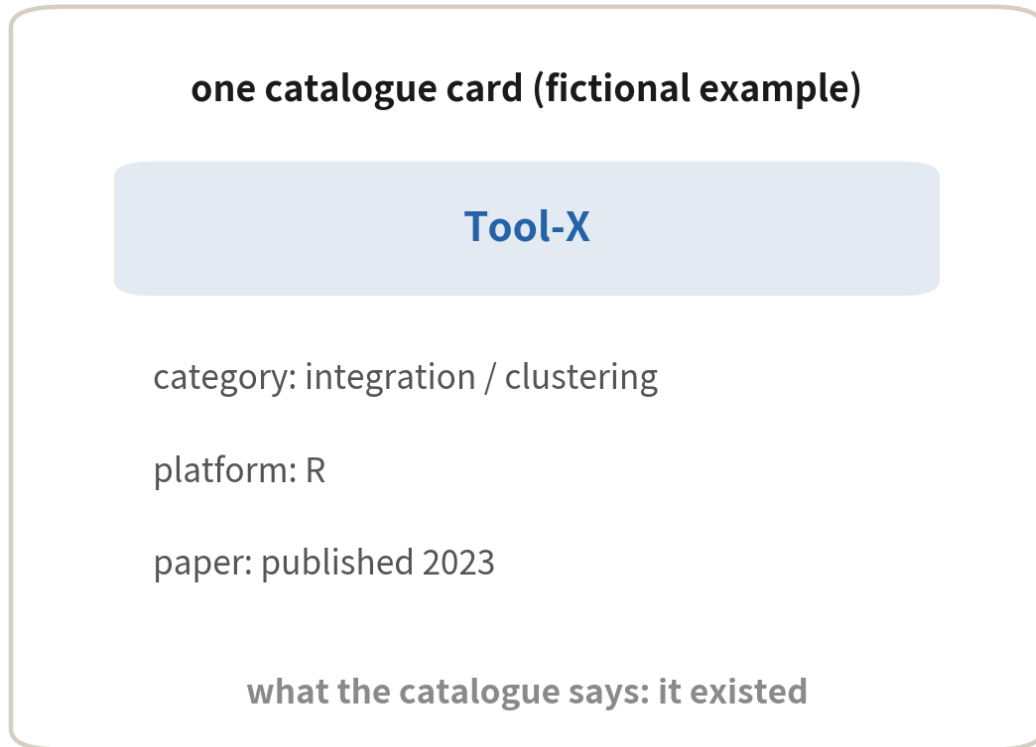
30+

task categories: QC, integration, trajectory, communication...

2024-05

since then, officially low-maintenance — the catalogue itself has slowed

Anatomy of a catalogue card: three things it won't tell you



three things no catalogue tells you

when was the last commit?

check GitHub yourself — a listed tool can be long dead

does it still install today?

the CRAN / PyPI / Bioconductor page settles it

any benchmark behind it?

catalogues do not compare performance — evidence lives elsewhere

a large share of the 1,847 listings is no longer maintained — listed is neither recommended nor alive

schematic

A listing only says 'it existed' — commits, installs, and benchmarks are on you to check

Trap 1: reading 'listed' as 'recommended'

What it is

Finding a tool on scRNA-tools with a category and a paper, and using it on that basis alone.



Why it happens

A catalogue's job is to list, not to vet — a large share of the 1,847 listings is no longer maintained.

What it costs you

Replay the cold open: three months after the pipeline is wired, the repo is archived, the install breaks, the analysis restarts.

THE RULE FOR THIS SECTION

Listed is not recommended.

Listed is not even alive.

Usability rests on three checks: the last commit, a formal release channel, benchmark backing.

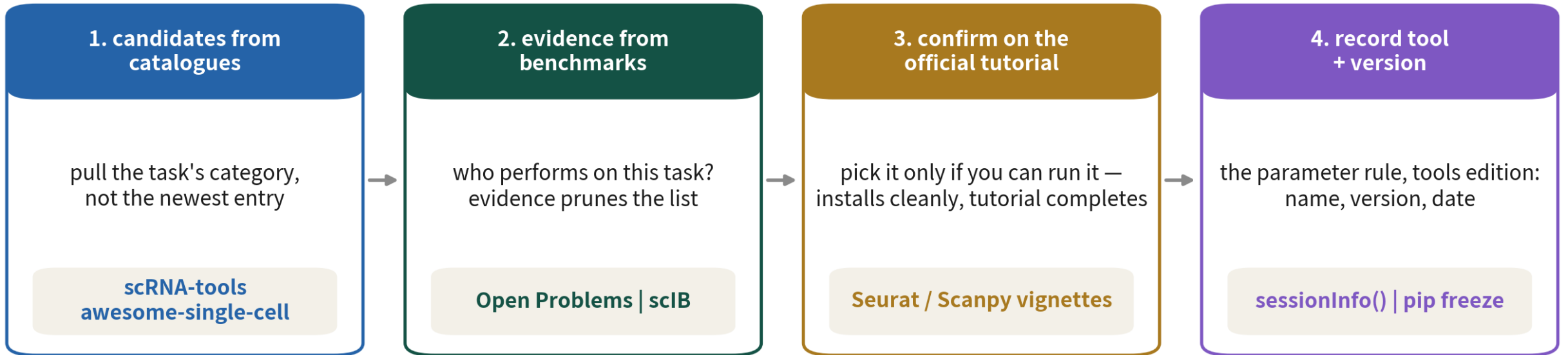
Section two builds them into a routine.

02

Choosing a tool in four steps

The core of this episode — replacing gut feel with a walkable routine

The four steps at a glance



the classic trap: grab 'the newest' at step one and jump straight past step four — skipping both middle steps

The classic trap skips both middle steps — grab 'the newest' at step one and start coding

The four steps, one by one

1. Catalogue

Pull the task's category on scRNA-tools; top up the newest from awesome-single-cell (still merging when we verified)

2. Benchmark

The Open Problems leaderboard, scIB, systematic comparisons — let evidence prune the list

3. Tutorial

It installs, and you can walk the official example yourself — the Seurat / Scanpy vignettes set the standard

4. Record

The parameter rule, tools edition: name, version, date — into the analysis notes

Step one: the catalogues

Scale and maintenance status as seen on the official pages, 2026-09

Catalogue	Scale (as seen)	When to use it
scRNA-tools.org	1,847 tools, 30+ categories	First stop for candidates by task; mind the low-maintenance notice
awesome-single-cell	3.8k stars, MIT, very active	The newest tools scRNA-tools hasn't listed yet
Bioconductor biocViews 'SingleCell'	Package count is a live page (copy the day's number)	Formally released R tools: tested, with vignettes
The scverse ecosystem list	12 core + 124 ecosystem	The Python-side bar for 'is this package legit'
Museum of Spatial Transcriptomics	1,098 spatial tools (compiled 2026-01)	Spatial has no scRNA-tools equivalent — this fills the role

Step two: the four evidence tiers

step two's four evidence tiers — search top-down, stop when you hit one

		when to use it
textbook chapter	sc-best-practices chapters: textbook-grade advice with cited evidence	when you want a defensible default
living leaderboard	Open Problems: 13 tasks • 125 methods • 94 gold-standard datasets	for 'how do the newest methods do' — fresher than any single paper
task-specific benchmark	scIB (Nat Methods 2022): the standard for integration; you can run the metrics yourself	choosing integration methods; sensitivity analyses
single benchmark papers	PubMed: '<task> benchmark single-cell'; prefer Nat Methods / Genome Biology	when the task is outside the 13 (communication, CNV, doublets...)

all four tiers share one caveat: check whether the benchmark's datasets resemble your setting — a benchmark's scope is a hypothesis too

Search top-down and stop when you hit one; the shared caveat — do the benchmark's datasets resemble yours?

Open Problems: the living leaderboard

CZI-funded, NumFOCUS fiscal host; a 2025 Nat Biotechnol platform paper

13

open problems (tasks)

125

methods, head to head

94

gold-standard datasets

Steps three and four: confirm and record

It installs



Check the release page on CRAN / PyPI / Bioconductor for the last release; removal to the CRAN archive is a loud warning

The example completes



Run the official tutorial end to end yourself — a tool you can't run is not your tool, however strong

The version gets recorded



Tool, version, date go into the analysis notes — the same discipline as a QC threshold

Step four as two commands

```
# R 端：版本快照存檔（B1 的 sessionInfo() 習慣，工具版）
writeLines(capture.output(sessionInfo()),
           "logs/sessionInfo_2026-09-07.txt")

# Python 端（終端機）：
# pip freeze > logs/requirements_2026-09-07.txt
```

Date in the filename — tool, version, date captured at once; when a reviewer asks, open the file

The three survival checks

1. date of the last commit

visible at a glance on GitHub;
an archived (read-only) repo = red light

2. a formal release channel

a formal release on Bioconductor /
CRAN / PyPI; CRAN-archived = loud warning;
Bioconductor: build report must be green

3. benchmark backing

Open Problems / scIB / a systematic
comparison — someone has measured it
on a standard task

the bar: at least two of the three — missing two or more, switch tools

Last commit, formal release channel, benchmark backing — at least two of three

Trap 2: copying the newest tool

What it is

Seeing a new method on social media or in a paper and swapping it in — no benchmark, no tutorial run.



Why it happens

New is not better: most new tools have entered no benchmark and survived nobody else's data.

What it costs you

Skipping both middle steps is the classic trap — no evidence read, no run verified, every problem deferred to you, three months from now.

CORE IDEA

**Catalogue for candidates, benchmark for evidence,
tutorial to confirm, version on record.**

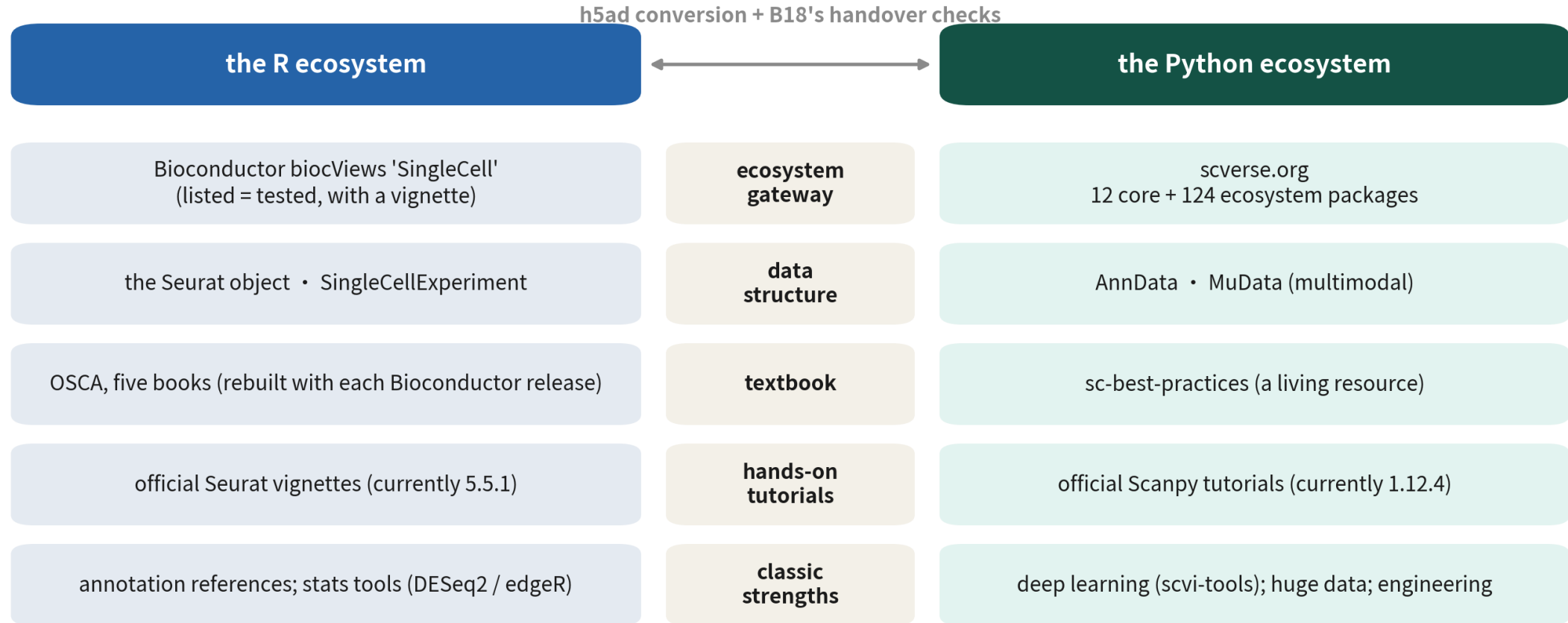
For tasks this course covered, start with the course's tool — switch only when it fails you, and walk the four steps when you do.

03

The two ecosystems

R and Python are two mirrored ecosystems, not two worlds (B18)

Task by task: R versus Python



every row has a counterpart on the other side — two mirrored ecosystems, not two worlds (B18)

Every row has a counterpart across the aisle — the bridge is h5ad conversion plus B18's handover checks

The twelve scverse core packages

As seen on scverse.org; the 124-package ecosystem list implies a baseline quality bar

Role	Packages	Note
Data structures	AnnData • MuData • SpatialData	Unimodal, multimodal, spatial — B18's h5ad is AnnData on disk
General analysis	Scanpy • Muon • Squidpy • scvi-tools	The main pipeline, multimodal, spatial, deep-learning models
Specialised tasks	Scirpy • SnapATAC2 • rapids-singlecell • Pertpy • decoupler	Immune receptors, chromatin, GPU acceleration, perturbations, activity inference

Current versions, into your notes

Verified 2026-09; the Scanpy docs now live at scanpy.scverse.org

5.5.1

current Seurat
(satijalab.org/seurat)

1.12.4

current Scanpy

12 + 124

scverse core + ecosystem
packages

The R route versus the Python route

Keep your main line where you're fluent; treat the other side as a toolbox (B18's advice)

the R route

Seurat + Bioconductor

- textbook: OSCA, five books, rebuilt each Bioconductor release
- strengths: annotation references; stats tools (DESeq2 / edgeR)
- packages via biocViews — listed = tested, with a vignette

the Python route

The scverse family

- textbook: sc-best-practices, chapters with benchmark evidence
- strengths: deep learning (scvi-tools), huge data, engineering
- packages via the ecosystem list — a baseline quality bar

Crossing over: three gates + one spot check

three gates + one spot check (B18's checklist, condensed)

gate 1: shape orientation

n_obs = cells, n_vars = genes —
flip it and everything flips

gate 2: obs columns

QC columns, clusters, cell types
all present, right dtypes

gate 3: obsm embeddings

are X_pca and X_umap there?

one spot check

pick one cell, one gene,
print on both sides — must match

dropped by default — a to-do list, not a disaster:

- scale.data → recompute with sc.pp.scale
- neighbour graph → rebuild with sc.pp.neighbors (and pin the parameters)
- extra assays like ADT → export separately, or switch to MuData

the handover note must say what was left behind — losing things is fine; not knowing is not (B18)

B18's checklist, condensed — what drops is a to-do list, not a disaster; say in the handover note what stayed behind

THE RULE FOR THIS SECTION

Two mirrored ecosystems — not two worlds.

Every task has a counterpart on the other side; treat every crossing as a house move, with an inspection (B18).

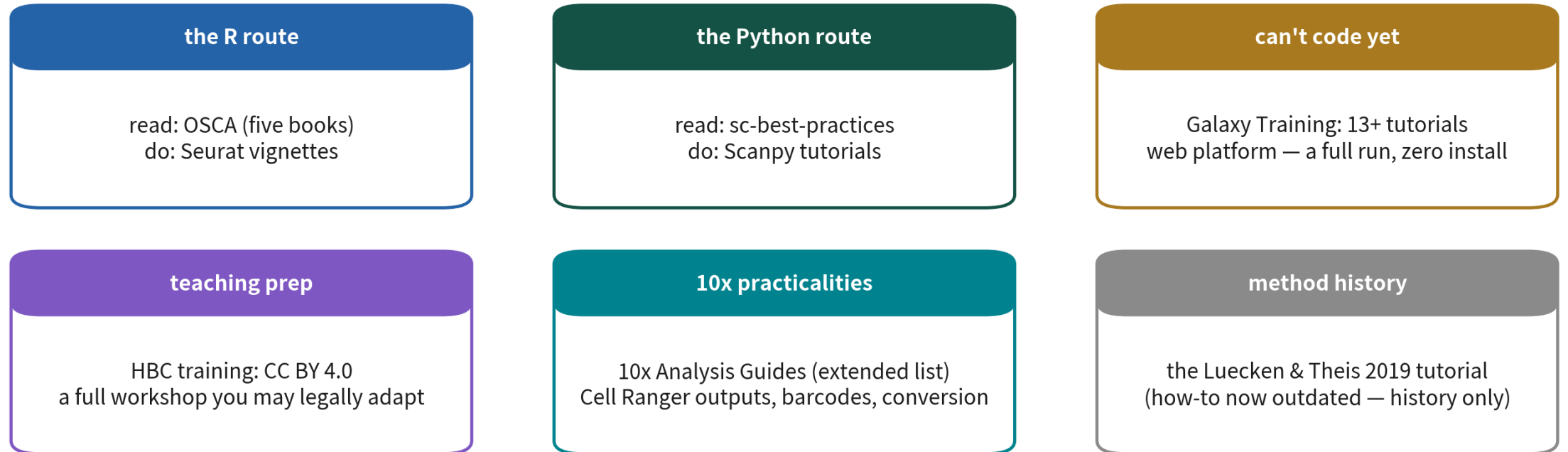
04

Where to learn

Books, courses, official tutorials — and one cautionary tale that died

The learning map: six situations

six situations, six entry points — all verified alive 2026-09 (the archived one is not on this map)



OSCA and sc-best-practices are living materials — record the date you accessed them

All verified alive 2026-09; find your situation, enter through that cell

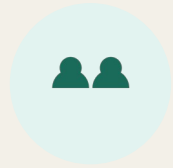
The four core materials, as of now

Status as seen 2026-09; for living materials, record your access date

Material	Status (as seen)	Role
OSCA	Five books; the 3.21 edition built 2025-06 (Amezquita, Nat Methods 2020)	The R / Bioconductor route's textbook
sc-best-practices	Apache 2.0, a living resource (Heumos, Nat Rev Genet 2023)	The Python route's textbook; the citation for 'why this method'
Seurat vignettes	Currently 5.5.1; from intro to spatial	The official counterpart of this course's pipeline
Scanpy tutorials	Currently 1.12.4; docs moved to scanpy.scverse.org	The other half of B18's cross-ecosystem rerun

Courses and practice: three entrances

HBC training



A full scRNA-seq workshop under CC BY 4.0 — teachers may legally adapt it; the licence says so in plain text

Galaxy Training



13+ tutorials, browser-based, zero install — the lowest bar for a first complete run before you can code

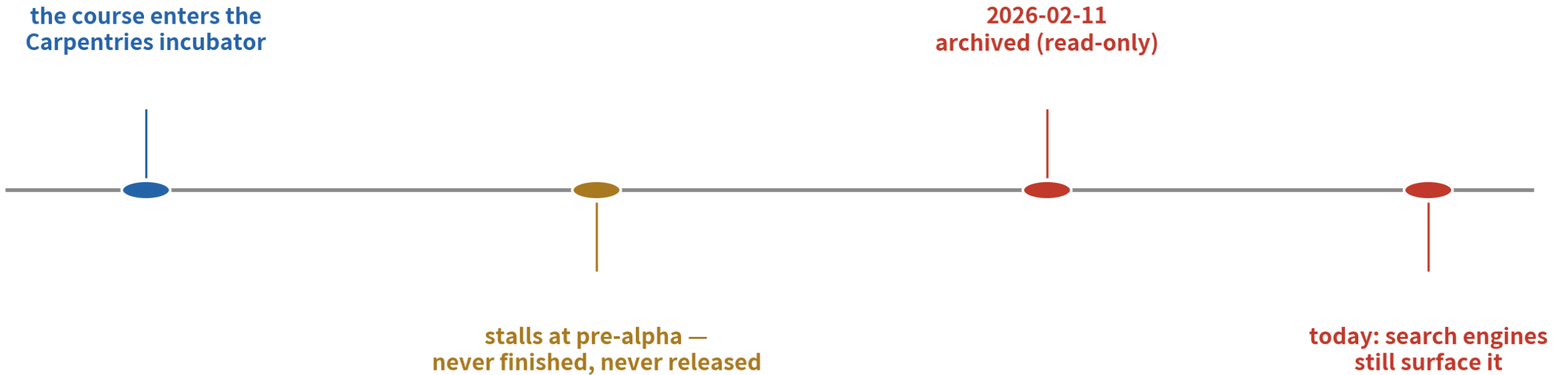
10x Analysis Guides



From the extended list: the official take on Cell Ranger outputs, barcode structure, conversion

A cautionary tale: the lifeline of one course

the lifeline of one course (spacing schematic; the archive date is verified)



teaching materials need the survival check too — for free Carpentries-style courses, use HBC or Galaxy Training instead

schematic

The Carpentries incubator scRNA-seq course stalled at pre-alpha and was archived 2026-02-11 — search engines still surface it

Trap 3: page one of search as a live recommendation

What it is

Searching 'scRNA-seq course' and recommending page one to students — the archived Carpentries course included.



Why it happens

Search rank reflects historical links, not maintenance; an archived repo's pages still look intact.

What it costs you

Students follow a dead pre-alpha lesson and no fix ever comes — point them to HBC or Galaxy Training instead.

Uncovered tasks: the benchmark-paper search

The query



PubMed: '<task>
benchmark single-cell'
— for tasks outside the
13, like
communication, CNV,
doublets

What to read first



Systematic
comparisons in Nat
Methods / Genome
Biology — not a single
tool's own paper

How to read it



Check whether the
benchmark's datasets
resemble your setting
— a benchmark's
scope is a hypothesis
too

THE RULE FOR THIS SECTION

**Date the living materials.
Stop recommending the dead ones.**

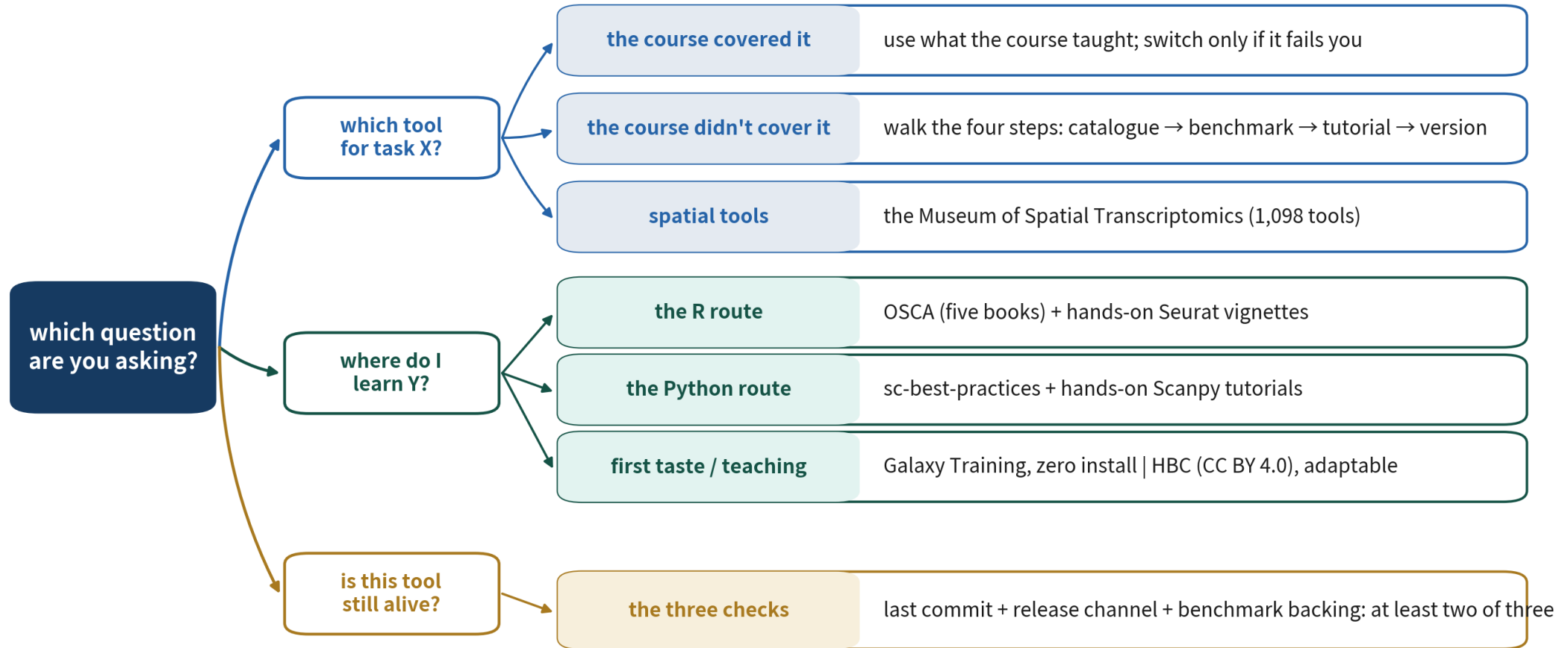
OSCA and sc-best-practices update continuously — record your access date; materials face the same survival check as tools.

05

The method, on one page

Three questions, one decision tree — walk it from now on

The tool-choice decision tree



whichever leaf you land on, the last step is the same: record the tool, the version, the date

Whichever leaf you land on, the last step is the same: record the tool, the version, the date

The three questions, at a glance

Your question	The answer
Which tool for task X?	Course covered it → use that; not covered → walk the four steps; spatial → Museum of ST
Where do I learn Y?	R → OSCA + Seurat vignettes; Python → sc-best-practices + Scanpy; no code → Galaxy; teaching → HBC
Is this tool still alive?	Last commit + formal release channel + benchmark backing — at least two of three

ONE-LINE SUMMARY

**Tools go stale.
The four steps do not.**

B18 said it: tools get replaced, you remain. This episode equips that 'you' — a method that survives every tool swap.

Check yourself 1

A tool is listed on scRNA-tools. What does that establish?

- A It passed an official quality review
- B It is still maintained
- C It existed — usability rests on three separate checks
- D It performs well on benchmarks

Answer C. A catalogue lists; it does not vet — a large share of the 1,847 entries is unmaintained. Usability needs the last commit, a formal release channel, and benchmark backing — at least two of three.

Check yourself 2

You want to swap the integration method from B11. Where is the standard evidence for step two?

- A Pick the most GitHub stars
- B The sclB benchmark (Luecken, Nat Methods 2022) — you can even run its metrics
- C The comparison in the tool's own paper
- D Majority opinion on a forum

Answer B. Integration has its own standard benchmark: sclB. Beyond the paper's conclusions, its package (v1.2.1, still releasing 2026-04) lets you recompute the metrics on your own data — exactly how the advanced exercises run sensitivity analyses.

Check yourself 3

A student who cannot code yet wants one full scRNA-seq run. Which entrance do you recommend?

- A The Carpentries incubator scRNA-seq course
- B Read all five OSCA books first
- C Galaxy Training — browser-based, zero install
- D Install Seurat and type along with a vignette

Answer C. Galaxy Training has 13+ tutorials on a web platform with zero install — the lowest bar for a first complete run. A is the trap: that course stalled at pre-alpha and was archived 2026-02-11 — still searchable, no longer recommendable.

NEXT EPISODE

BS4: A guide to annotation resources

One cluster, three tools, three different names — which is right? Marker dictionaries, references, atlases, tumour tools:

next time we walk the annotation resource map. See you there.